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February 24, 2026

Attn: Document Control Desk
U. S. Nuclear Regulatory Commission
Washington, DC 20555-0001

10 CFR 50.73

**SUSQUEHANNA STEAM ELECTRIC STATION
LICENSEE EVENT REPORT 50-388/2025-001-01
UNIT 2 LICENSE NO. NPF-22
PLA-8214**

Docket No. 50-388

Attached is Licensee Event Report (LER) 50-388/2025-001-01. The LER reports an event involving the trip of the Unit 2 main generator and subsequent automatic reactor scram that was determined to be reportable in accordance with 10 CFR 50.72(b)(2)(iv)(B) and 10 CFR 50.72(b)(2)(xi). This event is also reportable in accordance with 10 CFR 50.73(a)(2)(iv)(A) as an event that resulted in an automatic actuation of the Reactor Protection System (including a reactor scram).

There were no actual consequences to the health and safety of the public as a result of this event.

This letter contains no new or revised regulatory commitments.

A handwritten signature in black ink, appearing to read "Mark Jones", with a horizontal line extending from the end of the signature.

M. Jones

Attachment: LER 50-388/2025-001-01

Copy: NRC Region I
Mr. R. Wehrmann, NRC Senior Resident Inspector
Mr. R. Guzman, NRC Project Manager
Mr. M. Shields, PA DEP/BRP

Attachment to PLA-8214

LER 50-388/2025-001-01



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form)

<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name Susquehanna Steam Electric Station (SSES) Unit 2	☒ 050	2. Docket Number 00388	3. Page 1 OF 3
	☐ 052		

4. Title Arc Flash on 500 kilovolt Transmission Lines Resulted in Main Generator Trip and Automatic Reactor Scram
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5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☐ 050	Docket Number
10	28	2025	2025	- 001 -	01	02	24	2026	Facility Name	☐ 052	Docket Number

9. Operating Mode 1	10. Power Level 100
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11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☒ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☒ 10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☒ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	☒ 10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	☒ 10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		
☐ OTHER (Specify here, in abstract, or NRC 366A).					

12. Licensee Contact for this LER

Licensee Contact Jonathan Youells, Nuclear Regulatory Affairs	Phone Number (Include area code) 570-542-1461
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13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS

14. Supplemental Report Expected

☒ No	☐ Yes (If yes, complete 15. Expected Submission Date)	15. Expected Submission Date	Month	Day	Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On October 28, 2025, at approximately 19:04, Susquehanna Steam Electric Station (SSES) Unit 2 experienced a main turbine trip followed by an automatic reactor scram when an arc flash on the 500 kilovolt (kV) lines occurred during a hydrogen vendor manual venting evolution. The event was reported by Event Notification 58012 in accordance with 10 CFR 50.72(b)(2)(iv)(B) as an event that resulted in an automatic actuation of the Reactor Protection System and 10 CFR 50.72(b)(2)(xi) as an event that resulted in an offsite notification made to the Pennsylvania Emergency Management Agency. This event is also reportable in accordance with 10 CFR 50.73(a)(2)(iv)(A) as an event that resulted in an automatic actuation of the Reactor Protection System. The cause of the event was arc flash as a result of hydrogen gases vented from the vendor-owned Hydrogen Tank Farm during normal vendor evolution contacting overhead 500 kV transmission lines. Key corrective actions include establishment of an alternative location for venting the vendor hydrogen tank trailer, revision of venting evolution instructions, and evaluation and implementation of options to increase clearance between the Hydrogen Tank Farm vent and overhead transmission lines.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME

Susquehanna Steam Electric Station (SSES)
Unit 2

**050****052****2. DOCKET NUMBER**

00388

3. LER NUMBER**YEAR**

2025

SEQUENTIAL**NUMBER**

001

REV**NO.**

01

NARRATIVE**CONDITIONS PRIOR TO THE EVENT**

Unit 1: Mode 1, approximately 100% Rated Thermal Power
Unit 2: Mode 1, approximately 100% Rated Thermal Power

There were no inoperable structures, systems, or components at the start of the event that contributed to the event.

EVENT DESCRIPTION

On October 28, 2025, at approximately 19:04, Susquehanna Steam Electric Station (SSES) Unit 2 experienced a main turbine trip [EII System Code - IT] followed by an automatic reactor scram when an arc flash on the 500 kilovolt (kV) lines occurred during a hydrogen vendor manual venting evolution. The event was reported by Event Notification 58012 in accordance with 10 CFR 50.72(b)(2)(iv)(B) as an event that resulted in an automatic actuation of the Reactor Protection System and 10 CFR 50.72(b)(2)(xi) as an event that resulted in an offsite notification made to the Pennsylvania Emergency Management Agency. This event is also reportable in accordance with 10 CFR 50.73(a)(2)(iv)(A) as an event that resulted in an automatic actuation of the Reactor Protection System (including a reactor scram).

The following timeline demonstrates the sequence of events:

10/28/2025 Before 18:16: Vendor driver notifies Control Room that they will be filling the liquid hydrogen tank at the Hydrogen Tank Farm.

10/28/2025 18:16: Vendor driver begins filling the liquid hydrogen tank at the Hydrogen Tank Farm.

10/28/2025 19:04: Liquid hydrogen tank is filled. Vendor driver begins to depressurize the tanker truck via the hydrogen tank exhaust stack at the Hydrogen Tank Farm.

10/28/2025 19:04: An arc is visible in the area of the resultant plume from the hydrogen release on the 500 kV conductors overhead the hydrogen tank in the Hydrogen Tank Farm.

10/28/2025 19:04: Unit 2 experiences an automatic scram due to Main Turbine trip caused by Main Generator [EII System Code - TB] differential line current and enters Mode 3.

All plant systems responded normally following the automatic reactor scram. Decay heat was removed by discharging steam to the main condenser using the turbine bypass valves. Unit 1 was unaffected by the event and remained at approximately 100% Rated Thermal Power.

The affected 500 kV lines were placed in service in August 2023. While similar liquid hydrogen activities are routinely performed at this location, this type of event has not previously occurred.

Following visual inspection and testing of the 500kV lines and a review of the post-trip plant response, SSES Unit 2 was returned to Mode 1 operation on October 31, 2025.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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1. FACILITY NAME

Susquehanna Steam Electric Station (SSES)
Unit 2

**050****052****2. DOCKET NUMBER**

00388

3. LER NUMBER**YEAR**

2025

**SEQUENTIAL
NUMBER**

001

**REV
NO.**

01

NARRATIVE**CAUSE OF THE EVENT**

The cause of the event was arc flash as a result of hydrogen gases vented from the vendor-owned Hydrogen Tank Farm during normal vendor evolution contacting overhead 500 kV transmission lines. This cause was attributed to the modifications to install the 500 kV transmission lines to the 500 kV substation not adequately accounting for clearance limits between the transmission lines and the hydrogen exhaust stack in the Hydrogen Tank Farm.

ANALYSIS/SAFETY SIGNIFICANCE

The actual consequence of the event was the automatic scram of Unit 2 from approximately 100% power operation due to a Generator Load Reject and Main Turbine Trip. The scram was not complex; all safety systems operated as designed. Main Turbine Bypass Valves [EIS System Code - JI] and Safety Relief Valves [EIS Component Code - RV] opened automatically to maintain reactor vessel pressure. Reactor water level was stabilized and maintained with reactor feedwater [EIS System Code - SJ]. There was no Emergency Core Cooling System or Engineered Safeguard Function [EIS System Code - JE] initiations nor were any required.

The shutdown was reviewed regarding FSAR Chapter 15 and determined to be fully bounded by the analysis. Chapter 15.2.2 specifically addresses the Generator Load Rejection with the Steam Bypass System operable and classifies this as a non-limiting event. Main Turbine Bypass and Safety Relief Valves operated as designed during this event to mitigate reactor pressure excursion. All safety systems were observed to operate as designed.

The condition described herein did not result in a Safety System Functional Failure. Accordingly, this event will not be counted as a Safety System Functional Failure in the Reactor Oversight Process Performance Indicators. There were no actual consequences to the health and safety of the public as a result of this event.

CORRECTIVE ACTIONS

1. An alternative location for venting of the vendor hydrogen tank trailer has been specified and the vendor has accepted updated delivery instructions prohibiting intentional manual venting at the Hydrogen Tank Farm.
2. Identify and implement a solution that ensures clearances between the Hydrogen Tank Farm hydrogen exhaust stack and overhead transmission lines meet the identified clearance requirements.

COMPONENT FAILURE INFORMATION

N/A

PREVIOUS OCCURRENCES

None.